



PUCP – INSTITUTO DE RADIOASTRONOMÍA (INRAS)

INTISAT – Revisión Crítica de Diseño

Critical Design Review (CDR) – Modelo de Ingeniería

Departamento de Ingeniería – Pontificia Universidad Católica del Perú

Noviembre 2025

Índice de figuras

1.1	Vista del modelo 3D de INTISAT.	8
1.2	Logotipo oficial de la misión INTISAT.	10
3.1	Diagrama de bloques general de la arquitectura del OBC.	18
3.2	Flujogramas del modo de operación del satélite.	19

Índice de cuadros

2.1	Parámetros orbitales esperados de INTISAT.	12
2.2	Presupuesto preliminar de masa de INTISAT 2U. Valores TBD pendientes de medición del Modelo de Ingeniería.	13
2.3	Consumo de potencia por módulo y <i>duty cycle</i> . Fuente: Requerimiento de energía - INTISAT.xlsx	14
2.4	Link budget preliminar de INTISAT. Detalle de cálculo en Anexo. . . .	14
2.5	Resumen del uso de buses internos por subsistema (interfaz PC-104). .	14

Nomenclatura

ADCS	Attitude Determination and Control Subsystem
AIT	Assembly, Integration and Testing
BMS	Battery Management System
CCSDS	Consultative Committee for Space Data Systems
CDR	Critical Design Review
ConOps	Concept of Operations
DoD	Depth of Discharge
EMC	Electromagnetic Compatibility
EPS	Electrical Power System
FEA	Finite Element Analysis
FPM	Fourier Ptychographic Microscopy
INRAS	Instituto de Radioastronomía
LCL	Latch-up Current Limiter
LEO	Low Earth Orbit
MPPT	Maximum Power Point Tracking
OBC	On-Board Computer
OBDH	On-Board Data Handling
P-POD	Poly-PicoSatellite Orbital Deployer
PUCP	Pontificia Universidad Católica del Perú
RF	Radiofrecuencia
SEU	Single Event Upset
SoC	State of Charge
TCS	Thermal Control Subsystem
TTC	Telemetry, Tracking and Command
TVAC	Thermal Vacuum Chamber
UHF/VHF	Ultra/Very High Frequency
WBS	Work Breakdown Structure

Índice general

Nomenclatura	3
1 Introducción	8
1.1 Descripción de la Misión	8
1.2 Objetivos de la Misión	9
1.3 Concepto de Operaciones (ConOps)	9
1.4 Miembros del Proyecto	9
1.5 Logotipo de Misión	10
2 Arquitectura del Sistema	11
2.1 Visión general del sistema	11
2.1.1 Diagrama de Potencia	11
2.1.2 Diagrama de Flujo de Datos	11
2.2 Presupuestos Técnicos y Análisis de Misión	12
2.2.1 Parámetros de Órbita y Vida Útil	12
2.2.2 Presupuesto de Masa (Mass Budget)	12
2.2.3 Presupuesto de Potencia (Power Budget)	12
2.2.4 Cálculo de Enlace (Link Budget)	13
2.3 Documento de Control de Interfaces (ICD)	13
2.3.1 Interface Data Sheet y Form Factor	13
3 OBC – Computadora de a Bordo	15
3.1 Diseño de Hardware y Arquitectura Central (OBDH)	15
3.1.1 Esquema de Memorias y Almacenamiento	15
3.1.2 Monitoreo Interno y Sensores	15

3.2	Capa de Aislamiento e Interfaz de Comunicaciones	16
3.3	Máquina de Estados y Modos de Operación	16
3.3.1	Modo Inicio (Deployment y Vuelo Temprano)	16
3.3.2	Modo Rutina (Operación Nominal)	17
3.3.3	Modo de Seguridad (Safe Mode)	17
3.3.4	Modo Decomisión (End of Life)	17
4	EPS – Electrical Power System	20
4.1	Subsystem Overview	20
4.2	Power Generation (Solar Panels)	20
4.3	Storage and Capacity (Battery Module)	20
4.4	Conditioning, Distribution and Safety	21
5	ADCS – Attitude Determination and Control	22
5.1	Introduction to Orbital Attitude Control	22
5.2	Passive and Dynamic Attitude Scheme	22
5.2.1	Passive Magnetic Stabilization and Materials	22
5.2.2	Active Management and Sensor Modules	22
6	TTC – Telecommunications	24
6.1	Tracking, Telemetry and Command Module (TTC)	24
6.2	LEO Transmission Architecture	24
6.2.1	UHF/VHF Links and Segmentation	24
6.3	Antennas and Electrical Deployment Mechanisms	25
6.3.1	Fixed Architecture and Ballistic Retention	25
6.3.2	Active Activation Sequence	25
7	Payloads	26
7.1	Strategic Description of the Instruments	26
7.2	Computational CMOS Microscopy	26
7.2.1	General Architecture	26
7.2.2	Optical Acquisition Block (<i>Lensless</i> Configuration)	26

7.2.3	Illumination Block	27
7.2.4	Validation Methodology and Metrology	27
7.3	Data Control and Processing	27
7.4	AI Acceleration On-Orbit	27
8	Structure and Mechanics	28
8.1	Modular General Structural Design	28
8.2	Structural Design and Mass Requirements	28
8.2.1	Aerospace-Alloy Properties	28
8.2.2	Geometry Standardization and CG Tolerance	28
8.3	Finite Element Analysis (FEA) Simulation	29
9	Thermal Control	30
9.1	LEO Thermal Control Mechanisms	30
9.1.1	Passive Control (Optical Coatings and Mass)	30
9.1.2	Active Control (EPS and OBC Integration)	30
9.2	Thermal Test Campaign (AIT)	30
9.2.1	Thermal Cycling	31
9.2.2	Thermal Vacuum Test (Bake-Out LEO)	31
10	Ground Segment	32
10.1	INRAS Ground Operational Mission	32
10.2	Ground Station Architecture, Control and Monitoring	32
10.3	Logical Ground Operations Cycle (Ops)	32
11	Integration, Verification and Validation (AIT)	34
11.1	Overall Definition of the AIT Campaign	34
11.2	Assembly Instructions and FlatSat	34
11.2.1	FlatSat (Bench Tests)	34
11.2.2	Mechanical Integration Procedure	35
11.3	Physical and Environmental Validation Test Plan	35
11.3.1	Physical-Mechanical Tests	35

11.3.2	Space Environment and EMC Tests	36
11.3.3	Recommended AIT campaign sequence	36
12	Mission and Project Management	37
12.1	Systems Management and Milestones	37
12.2	Schedule, WBS and Coupled Risk Management	37
A	References	38
B	Annexes	40
B.1	Additional diagrams, tables and budgets	40

Capítulo 1

Introducción

1.1 Descripción de la Misión

INTISAT es una misión de demostración tecnológica en formato **CubeSat 2U** liderada por la Pontificia Universidad Católica del Perú (PUCP) a través del Instituto de Radioastronomía (INRAS). El objetivo principal es validar la integración de una carga útil compacta de microscopía por imagen para operación en órbita terrestre, desarrollada por estudiantes e investigadores como parte de los esfuerzos de PUCP en educación y desarrollo de tecnología espacial.

INTISAT demostrará una plataforma totalmente integrada compuesta por una computadora de a bordo (OBC), un sistema de energía eléctrica (EPS), un sistema de telemetría, rastreo y comando (TTC), componentes estructurales y una carga útil científica. Todo el sistema está contenido en una envoltura mecánica CubeSat 2U. La implementación actual corresponde a un *Modelo de Ingeniería*, base para mejoras posteriores y una eventual calificación de vuelo.

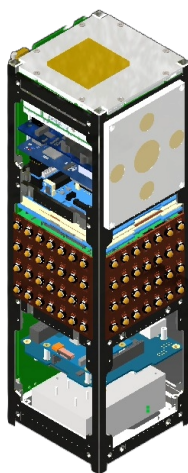


Figura 1.1: Vista del modelo 3D de INTISAT.

1.2 Objetivos de la Misión

Los objetivos principales de INTISAT son:

- Validar la integración y operación de un sensor de imagen miniaturizado para microscopía en órbita terrestre baja (LEO).
- Demostrar la funcionalidad de subsistemas desarrollados por estudiantes operando dentro de una plataforma de nanosatélite unificada.
- Desarrollar y verificar una arquitectura CubeSat escalable y modular alineada con estándares académicos y educativos.
- Servir como plataforma de entrenamiento práctico en ingeniería de sistemas, diseño, integración y pruebas de satélites.

1.3 Concepto de Operaciones (ConOps)

Siguiendo enfoques estandarizados de operación satelital, el ciclo operativo de INTISAT se define mediante cuatro modos de misión claramente delimitados. La gestión y transiciones entre modos las ejecuta la OBC (ver Cap. 3).

- **Modo Despliegue e Inicialización:** durante el lanzamiento todos los sistemas están físicamente deshabilitados por *kill-switches*. Tras la eyección, los sistemas se energizan, la OBC arranca, se ejecutan verificaciones iniciales de *housekeeping* y se inicia la secuencia de despliegue de antenas después de una ventana de silencio obligatoria de 40 minutos.
- **Modo Nominal (Rutina):** estado operativo estándar en el que el EPS maximiza la recolección solar, la baliza transmite periódicamente y la carga útil (Microscopía / IA) se activa durante ventanas predeterminadas sobre el territorio peruano u otros objetivos planificados.
- **Modo Contingencia (Seguridad):** estado seguro al que se entra automáticamente si se detecta una anomalía crítica (por ejemplo, SoC bajo de batería o eventos *latch-up*). Las cargas no esenciales se deshabilitan hasta la intervención del control terreno.
- **Modo Decomisión:** estado de fin de vida activado mediante telecomando desde tierra o por temporizador programado. Asegura la pasivación segura de la batería y el apagado de transceptores antes del reingreso atmosférico.

1.4 Miembros del Proyecto

INTISAT es una iniciativa académica impulsada por equipos interdisciplinarios de PUCP. La estructura del proyecto incluye:

- **INRAS – PUCP:** coordinación de misión, ingeniería de sistemas, desarrollo del módulo TTC e integración de la plataforma.
- **Facultad de Ciencias e Ingeniería – PUCP:** desarrollo de software, electrónica y subsistemas mecánicos.
- **Socios externos (fase futura):** colaboradores para pruebas, coordinación de lanzamiento u operaciones de misión.

1.5 Logotipo de Misión



Figura 1.2: Logotipo oficial de la misión INTISAT.

Capítulo 2

Arquitectura del Sistema

2.1 Visión general del sistema

El sistema del nanosatélite INTISAT está compuesto por un conjunto de submódulos interconectados sobre un bus estándar de facto (tipo PC-104 modificado o equivalente de la industria aeroespacial). La misión principal se asegura mediante un módulo de servicio (Bus) robusto frente a fallos y compatible con las distintas cargas útiles (cámara de microscopía CMOS, aceleración IA y otras planificadas).

2.1.1 Diagrama de Potencia

El módulo EPS (*Electrical Power System*) gestiona de forma centralizada todos los buses de poder, alimentando los subsistemas mediante rieles de 3,3 V, 5 V y la línea no regulada V_{batt} . Esto permite control estricto sobre los convertidores y habilita esquemas de encendido/apagado de módulos de manera individual, esencial para gestionar el consumo durante eclipses o ante eventos de *latch-up*.

2.1.2 Diagrama de Flujo de Datos

La ruta de datos tiene como nodo de control y mediación a la computadora de a bordo (OBC). La comunicación interna entre módulos del Bus se establece por protocolos estándar robustos de baja velocidad (I2C, SPI, UART) y un bus CAN para telemetría cruzada con subsistemas vitales (EPS). La OBC recolecta datos internos (*housekeeping*), telemetría general y datos del *payload*, los empaca en estructuras de enlace CCSDS/AX.25 y los enruta al segmento terreno (AYNISAT) a través del módulo transceptor UHF.

2.2 Presupuestos Técnicos y Análisis de Misión

Esta sección centraliza los presupuestos críticos para el ciclo de vida del nanosatélite, garantizando que cumple con los requerimientos restrictivos del formato 2U. Los datos numéricos se gestionan en `03_recursos/presupuestos/Requerimiento de energía - INTISAT.xlsx` y se consolidan aquí para el CDR.

2.2.1 Parámetros de Órbita y Vida Útil

Cuadro 2.1: Parámetros orbitales esperados de INTISAT.

Parámetro	Valor
Altitud orbital nominal	TBD km (LEO típica: 400–550 km)
Inclinación	TBD° (típica SSO: 97–98°)
Periodo orbital	TBD min (~93 min para 500 km)
Tiempo de vida en órbita	TBD años (lifetime analysis pendiente)
Estación terrena de referencia	INRAS (PUCP, Lima, Perú)
Inyector previsto	P-POD (TBD: SpaceX, ISRO, JAXA, etc.)

2.2.2 Presupuesto de Masa (Mass Budget)

Según la especificación *CubeSat Design Specification* (CDS Rev 14, Cal Poly), la masa máxima de cada unidad debe ser $\leq 1,33$ kg. Al ser INTISAT un CubeSat 2U, la masa máxima permitida es 2,66 kg. La Tabla 2.2 resume el presupuesto preliminar.

Criterio de aceptación. Margen de masa ≥ 15 % sobre el máximo 2U (referencia FloripaSat-2: 16 %).

2.2.3 Presupuesto de Potencia (Power Budget)

El presupuesto de potencia sigue tres pasos: (i) preparar el presupuesto operacional, (ii) dimensionar la batería (DoD, ciclos), (iii) estimar degradación.

Dimensionamiento de batería. La profundidad de descarga (DoD) se calcula como:

$$\text{DoD} = \frac{D_i \cdot D_t}{C} \cdot 100 [\%] \quad (2.1)$$

donde D_i es la corriente de descarga, D_t el periodo de descarga, y C la capacidad de la batería.

Criterio: $\text{DoD} \leq 25$ % para vida útil LEO de 5 años. Referencia FloripaSat-2: $\text{DoD} \approx 8,5$ % con margen energético positivo (+643 mWh/órbita).

Cuadro 2.2: Presupuesto preliminar de masa de INTISAT 2U. Valores TBD pendientes de medición del Modelo de Ingeniería.

Subsistema	Modelo	NotasMasa [g]	
OBC (OBDH)	PUCP-INRAS OBC-STM32H7	TBC	TBD
TTC (UHF)	PUCP-INRAS UHF v1.0	TBC	TBD
EPS	PUCP-INRAS EPS v1.0	TBC	TBD
Batería	4× 18650 Li-Ion	TBC	TBD
Antena	Dipolos UHF plegados	TBC	TBD
ADCS (pasivo)	Imán + barras histéresis	TBC	TBD
Payload Microscopía CMOS	OV5647 + OLED + RPi	TBC	TBD
Payload IA (planificado)	—	—	—
Interfaz PC-104	PUCP-INRAS	TBC	TBD
Paneles solares	Triple-junction COTS	TBC	TBD
Estructura	Al-7075-T73, 2U	TBC	TBD
Cables (harness)	—	TBC	TBD
Otros	Tornillería, marcado	TBC	TBD
Total	—	—	TBD
Máximo CubeSat 2U	—	—	2660
Margen	—	—	TBD

2.2.4 Cálculo de Enlace (Link Budget)

El presupuesto de enlace se calcula para los enlaces uplink y downlink en banda UHF, considerando el peor caso orbital (satélite en horizonte, $\theta = 0^\circ$).

Criterio: margen ≥ 3 dB. Referencia FloripaSat-2: downlink $\geq 9,57$ dB, uplink $\geq 17,81$ dB.

2.3 Documento de Control de Interfaces (ICD)

2.3.1 Interface Data Sheet y Form Factor

Las placas de los subsistemas se interconectan mediante el bus PC-104 modificado. La distribución de pines, voltajes y señales de control se detalla a continuación. Cada subsistema declara su uso de rieles de potencia, buses de datos (I2C, SPI, UART, CAN) y GPIO compartidos.

El detalle pin-a-pin del conector PC-104 y los ICDs eléctricos completos están en los respectivos CDRs de subsistema (01_CDRs_subsistemas/).

Cuadro 2.3: Consumo de potencia por módulo y *duty cycle*. Fuente: Requerimiento de energía - INTISAT.xlsx.

Módulo	Duty Cycle [%]	Potencia [mW]
OBC (riel 3,3 V)	100	4060
OBC (riel 5 V, CANBUS)	100	1100
TTC UHF (RX)	95	TBD
TTC UHF (TX, beacon)	5	TBD
EPS (idle)	100	TBD
Batería (heater, eclipse)	10	TBD
Antena (despliegue, una vez)	0	TBD
Payload Microscopía CMOS	TBD	TBD
Payload IA / SAI (planificado)	0	—
Total satélite	—	TBD

Cuadro 2.4: Link budget preliminar de INTISAT. Detalle de cálculo en Anexo.

Parámetro	Uplink	Downlink	Unidad
Frecuencia	TBD	TBD	MHz
Potencia de transmisión	TBD	TBD	dBm
Ganancia de antena TX	TBD	TBD	dBi
EIRP	TBD	TBD	dBm
Pérdida en espacio libre (FSPL)	TBD	TBD	dB
Pérdida por polarización	TBD	TBD	dB
Ganancia de antena RX	TBD	TBD	dBi
G/T	TBD	TBD	dB/K
C/N ₀ recibido	TBD	TBD	dB·Hz
E _b /N ₀ disponible	TBD	TBD	dB
Margen	TBD	TBD	dB

Cuadro 2.5: Resumen del uso de buses internos por subsistema (interfaz PC-104).

Subsistema	Riel de potencia	Buses de datos
OBC	3,3 V + 5 V	I2C, SPI, UART×4, CAN, GPIO×22
EPS	V_{batt}	I2C (telemetría), CAN (housekeeping)
TTC UHF	5 V	SPI (módem), UART (control)
Payload Microscopía	3,3 V + 5 V (TBC)	I2C (OLED), CSI (cámara), UART (control)
Payload IA	TBD	TBD
ADCS pasivo	—	I2C (magnetómetro), GPIO (magnetorquers)

Capítulo 3

OBC – Computadora de a Bordo

3.1 Diseño de Hardware y Arquitectura Central (OBDH)

El hardware principal o módulo On-Board Data Handling (OBDH) es el *corazón* de la plataforma INTISAT, concebido de manera específica para coordinar las operaciones vitales, gestión de fallos y comunicaciones.

La arquitectura de procesamiento está basada en un microcontrolador de alto rendimiento **STM32H7 (144 pines)**, impulsado por osciladores de precisión (Cristal HSE de 24 MHz y LSE de 32.768 kHz). Este núcleo asegura el poder de cómputo necesario para tareas de tiempo real exigidas por las cargas útiles.

3.1.1 Esquema de Memorias y Almacenamiento

Para garantizar la persistencia de datos (Housekeeping y rutinas de Payload) y tolerar fallos inducidos por radiación mitigando eventos Single Event Upset (SEU), la OBC implementa un banco de memorias redundante operando a 3.3V:

- **Memorias de Alta Velocidad (x3):** Interconectadas mediante bus QSPI a 133 MHz, priorizando la ejecución rápida y buffers de comunicación.
- **Memorias de Baja Velocidad (x3):** Interconectadas mediante bus SPI tradicional a 25 MHz para redundancia y almacenamiento profundo no volátil.

3.1.2 Monitoreo Interno y Sensores

El acondicionamiento físico incluye un banco independiente de **6 unidades de sensores de temperatura** vinculados directamente al STM32H7 vía bus I2C, lo que permite trazar mapas de calor de la placa madre en tiempo real.

3.2 Capa de Aislamiento e Interfaz de Comunicaciones

Teniendo en cuenta el entorno disruptivo del espacio, la placa OBC no se comunica directamente con el bus estándar del satélite (PC/104). Se incorpora un diseño de aislamiento galvánico de señales:

- Una etapa de **Banco de Aisladores** protege los puertos expuestos hacia el conector PC/104. Esto involucra puertos críticos de Baja Velocidad (hasta 22 pines GPIO, 2x SPI, 2x I2C y 4x UART utilizados pesadamente por las cargas útiles).
- Se incluye un transceptor dedicado para los 2 canales del **Bus CAN** (CAN H/L), proporcionando inmunidad al ruido y un protocolo robusto para la telemetría cruzada con subsistemas vitales (como el EPS).

3.3 Máquina de Estados y Modos de Operación

La OBC gestiona el ciclo de vida de la misión mediante **cuatro modos de operación** consistentes con el ConOps definido en el Cap. 1: *Inicio*, *Rutina (Nominal)*, *Seguridad (Safe)* y *Decomisión*.

3.3.1 Modo Inicio (Deployment y Vuelo Temprano)

Al eyectarse del inyector P-POD, el *kill-switch* físico libera la energía proveniente del EPS y energiza el satélite. La OBC ingresa al modo de inicio ejecutando la siguiente rutina estricta:

1. **Temporizador de silencio:** ventana obligatoria de **40 minutos** sin ejecución de hardware crítico ni transmisiones de RF, permitiendo el alejamiento seguro del vehículo lanzador.
2. **Checklist de despliegue:** la OBC evalúa los siguientes parámetros:
 - Energía: consulta al EPS si la batería supera el 70 % de SoC.
 - Estabilidad orbital: consulta al ADCS para confirmar que las rotaciones residuales están dentro de tolerancia.
3. Confirmados los niveles, ejecuta el despliegue físico de antenas e ingresa al *Modo Rutina*. Si falla o los niveles no son óptimos, cae por contingencia al *Modo de Seguridad*.

3.3.2 Modo Rutina (Operación Nominal)

Estado operativo estándar. La OBC opera cíclicamente sobre dos frentes simultáneos:

- Envía periódicamente (cada órbita por defecto) los datos de telemetría de sanidad (*housekeeping*) de OBC, EPS y radio UHF.
- **Priorización de cargas útiles:** autoriza la ejecución de experimentos científicos (ej. adquisición en microscopía CMOS), almacena los datos transaccionales en el banco de memorias, los empaca vía protocolo interno y los enruta para transmisión hacia tierra.

3.3.3 Modo de Seguridad (Safe Mode)

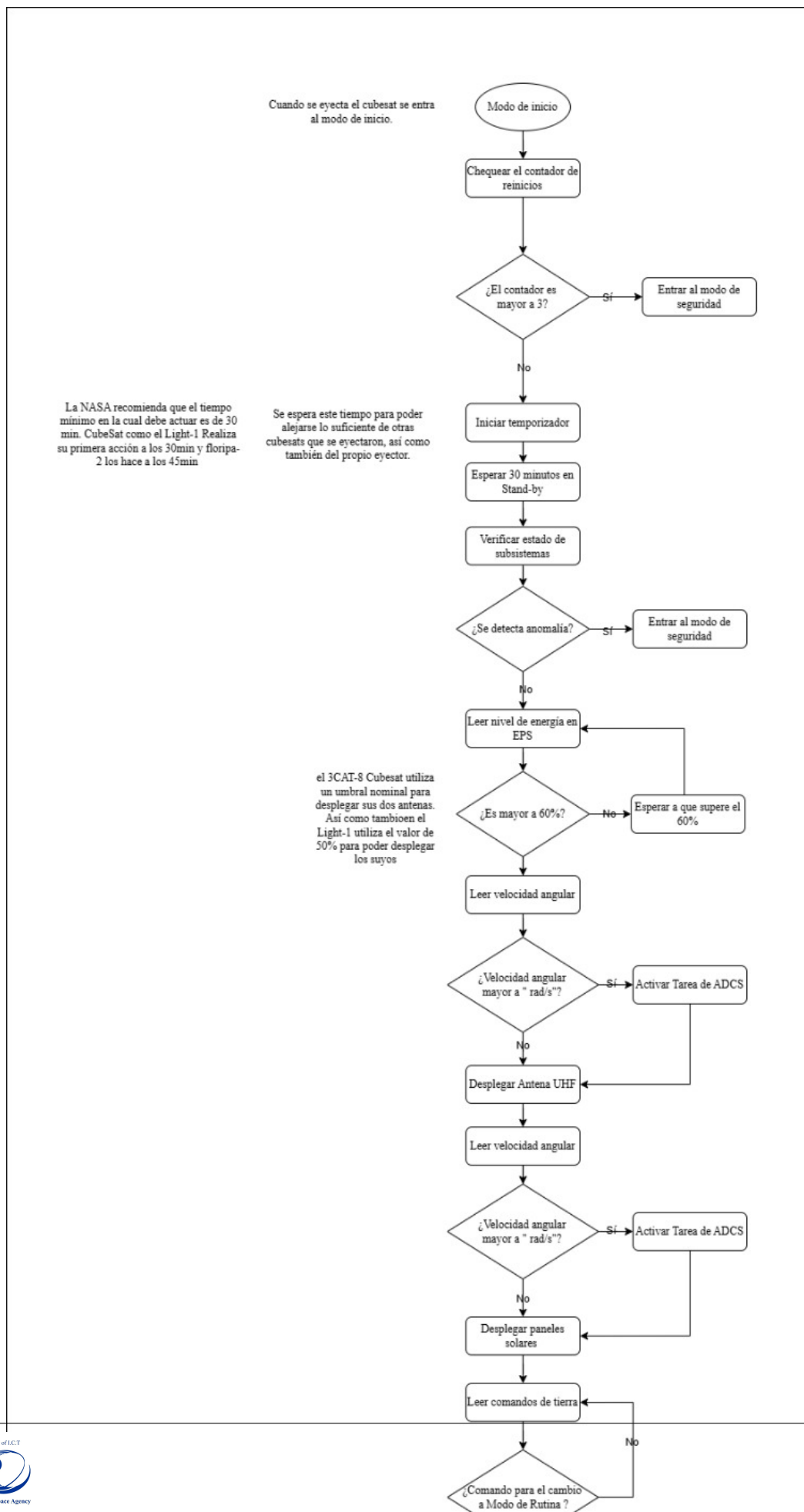
Ante el fracaso del *Modo Inicio* o un evento severo (sub-voltaje, SEU, latch-up), la máquina de estados suspende el Modo Rutina e ingresa al modo de supervivencia:

1. **Corte quirúrgico:** se apagan los puertos a subsistemas y *payloads* no vitales.
2. **Transmisión básica:** único envío periódico de telemetría housekeeping (EPS, OBC, UHF).
3. **Apuntamiento solar:** ordena al ADCS apuntar los paneles solares al sol de forma sostenida.
4. **Interbloqueo:** el satélite queda escuchando telecomandos críticos de tierra (HMAC/SHA) para inyectar correcciones asincrónicas (resets, forzar rutinas, salir del Safe Mode).

3.3.4 Modo Decomisión (End of Life)

Modo de fin de vida, activado mediante telecomando de tierra o por temporizador programado (lifetime expirado). Asegura el cese controlado de operación:

1. **Pasivación de batería:** descarga controlada y desconexión de las celdas para prevenir riesgo de explosión durante reentrada.
2. **Apagado de transceptores:** cese definitivo de transmisiones UHF.
3. **Latencia mínima:** la OBC entra en *deep sleep*, esperando el reingreso atmosférico natural por arrastre.



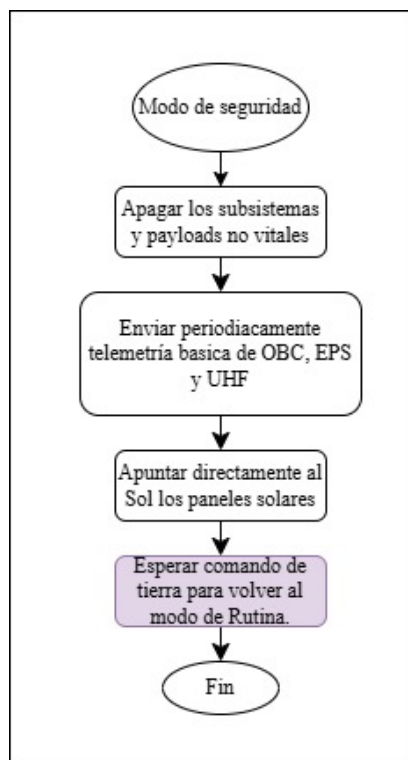


Figura 3.2: Flujogramas del modo de operación del satélite.

Capítulo 4

EPS – Electrical Power System

4.1 Subsystem Overview

The Electrical Power System (EPS) is responsible for three primary functions in LEO orbit: (i) energy harvesting via solar panels, (ii) storage via battery cells, and (iii) conditioning and safe distribution to the whole load bus. Additionally, the EPS indirectly orchestrates the life of the satellite by controlling the main switches (*kill-switches*).

4.2 Power Generation (Solar Panels)

INTISAT's energy source is provided by body-mounted arrays of solar cells (typically high-efficiency space-grade triple-junction photovoltaic cells, such as Emcore or AzurSpace) distributed across the external faces of the CubeSat prism.

- **Generation Management:** the EPS processor handles energy input through Maximum Power Point Tracking (MPPT) algorithms, ensuring maximum flow under highly variable irradiance conditions and optimizing panel efficiency.

4.3 Storage and Capacity (Battery Module)

Any excess energy harvested over immediate consumption during the sunlit phase is preferentially routed to the *Battery Module*.

- **Cells:** composed of robust Li-Ion cells in series/parallel (18650 form factor), with resistive thermal heaters to survive LEO orbital eclipses without freezing the electrolyte.
- **Battery Management System (BMS):** continuously measures Depth of Discharge (DoD), cell differential voltages, currents and thermistors, feeding the EPS module with warning parameters to prevent destructive under-voltages.

4.4 Conditioning, Distribution and Safety

Power is segmented from the battery and transformed via DC-DC converters into standard regulated lines (3.3 V and 5 V) and the native unregulated line (V_{batt}).

- **Analog protection and bus interface:** provides immediate hardware protection through Latch-up Current Limiters (LCL) that detect transient short-circuits induced by radiation or runaway payload consumption, cutting power on individual ports.
- **Integration (Interface Data Sheet):** the EPS communicates bidirectionally through I2C/CAN on PC-104, reporting sanity telemetry to the OBC and indicating the battery SoC, the central trigger for contingency routines.

Capítulo 5

ADCS – Attitude Determination and Control

5.1 Introduction to Orbital Attitude Control

The Attitude Determination and Control Subsystem (ADCS) aims to stabilize residual spins (*detumbling*) carried over from ejection and to lock a stable reference with respect to terrestrial vectors. Operating coupled with the OBC, it is critical for both the nominal mission phase (imaging pointing) and Safe Mode.

5.2 Passive and Dynamic Attitude Scheme

5.2.1 Passive Magnetic Stabilization and Materials

The first-level low-cost spin control is a thermally tolerant passive design:

- **Magnetic guide body:** one or more permanent magnets (e.g. N45 Neodymium) housed directly in the chassis or in a dedicated ADCS board, forcing the satellite to align and drag along the static magnetic field lines between Earth’s North/South poles.
- **Hysteresis bars:** perpendicular mechanical strips (e.g. *Permanorm*) inducing magnetic friction that damps and dissipates the CubeSat’s uncontrolled rotational energy (*detumbling*) by passively converting it to latent heat.

5.2.2 Active Management and Sensor Modules

For the precision required to continuously sun-point the solar panels and assist with microscopy and communications routing:

- **Determination:** multiple magnetometers on the Cartesian axes interact with LDR photodiodes sensing orbital solar fluxes. The matrix feeds the general ADCS measurement bus, reading the current orbital point with angular precision at high speed.



- **Actuators (dynamic):** a set of three or more *magnetorquers* (PCB-embedded coils with reversible polarization or copper cylinder tubes) generate artificial dipolar moments against the Earth's magnetic field, voluntarily modifying the attitude to channel light to the photovoltaic ports. They respond strictly in *Safe Mode* to direct OBC commands.

Capítulo 6

TTC – Telecommunications

6.1 Tracking, Telemetry and Command Module (TTC)

The TTC subsystem (*Telemetry, Tracking and Command*) establishes the unified RF link between the Ground Segment (INRAS station and affiliated networks) and the Space Segment synthesized by the OBC.

6.2 LEO Transmission Architecture

6.2.1 UHF/VHF Links and Segmentation

The internal modulation system assumes all transmissions are segmented under coupled radial subsystems:

- **Beacon (Life Beacon):** an independent hardware-driven transmitter periodically switched on, reporting vital compressed telemetry in CW Morse, AFSK or GMSK. It is the last barrier indicating life mode or restart without requiring full computation.
- **Uplink / Downlink UHF/VHF:** bidirectional half-duplex or full-duplex transit. The Payload transfers raw segmented files via CCSDS packets over the SPI bus to the modem on the downlink. The uplink receives corrective commands from ground, verifying CRC and a secure code (HMAC/SHA) to authorize alterations of the operational memory.

6.3 Antennas and Electrical Deployment Mechanisms

6.3.1 Fixed Architecture and Ballistic Retention

The antenna design (commonly dipoles/monopoles or crossed metallic ribbons for isotropic gain) requires absolute volumetric folding fitted to the 2U *form factor*.

- Antennas are initially anchored peripherally with pre-tensioned wires made of a nylon polymer, physically bonded by burner resistors adjacent to the external PCB structure.

6.3.2 Active Activation Sequence

Total responsibility for the release lies with the *Initialization Mode* commanded by the OBC. Once the **40-minute** silence window (Ch. 3) is over, the electrical system injects high currents over a short period ($\sim 5\text{--}10$ seconds), generating the thermal fusion of the nylon thread and allowing the dynamic deployment by elastic release to the orthogonal position of omnidirectional LEO RF coverage.

Capítulo 7

Payloads

7.1 Strategic Description of the Instruments

The INTISAT mission incorporates as its orbital evaluation core a set of payloads oriented to optical biometrics supported by Artificial Intelligence (AI). The goal is to demonstrate low-consumption computational microscopy and *Edge Computing*-style analysis under severe mass and power constraints in a CubeSat-class platform.

7.2 Computational CMOS Microscopy

7.2.1 General Architecture

The microscopy system is organized into four functional blocks: optical acquisition, illumination, control/processing and storage. It operates under a centralized architecture where a payload computer (based on ARM-class architectures, such as Raspberry Pi 5) coordinates capture, illumination control and metadata persistence, ensuring experimental traceability.

7.2.2 Optical Acquisition Block (*Lensless* Configuration)

This block uses a commercial CMOS sensor operating in lensless / contact microscopy configuration. In this architecture, the biological or biomimetic sample is placed in close contact with the sensor plane, eliminating refractive optical elements. The digital image is formed from the direct recording of the spatial intensity distribution (diffraction/shadow pattern) produced by the interaction of incident light with the sample. Manual control of exposure time and analog gain is prioritized to maintain strict calibration and metric consistency.

7.2.3 Illumination Block

Implemented as a flat, programmable system, this block generates patterns controlled in intensity and geometry (uniform illumination to maximize contrast, or structured grid/point patterns) ensuring temporal stability.

7.2.4 Validation Methodology and Metrology

Due to the COTS nature of the system and the absence of factory calibration, spatial scale ($\mu\text{m}/\text{pixel}$) validation rests on comparative observation of standard objects, specifically polystyrene microspheres of nominal diameter $2\ \mu\text{m}$. Before operational flight, samples are cross-checked using calibrated laboratory optical microscopy and scanning electron microscopy (SEM), bounding errors caused by the diffraction limits of the lensless approach.

7.3 Data Control and Processing

The payload computer acts as the logical core of the system. Its role is limited to sensor initialization (CSI/I2C), manual exposure application, capture of basic saturation metrics, and pairing of raw images with their operational metadata (*Sample Tag*, *ExposureTime*, *Timestamp*).

7.4 AI Acceleration On-Orbit

The neural-network acceleration sub-block induces higher mathematical resolution or *Super-Resolution* algorithms on the diffractive captures of the microscopy block without saturating the radio link. This implementation allows extracting morphology and inferring key vector parameters, reducing the UHF link bottleneck and transmitting analyzed, condensed information to the ground segment.

Capítulo 8

Structure and Mechanics

8.1 Modular General Structural Design

The shell and fuselage make up the modular master container (standard CubeSat rigid stacking layout in 2U format). Its purposes are twofold: resistance to the mechanical stresses (compression and vibration) of leaving Earth, and providing precise rigid rails allowing high-speed injection into vacuum from the integrated launcher dispenser.

8.2 Structural Design and Mass Requirements

8.2.1 Aerospace-Alloy Properties

At the metal-mechanical level, vertical rails and side panels (Ribs and Side Panels) are machined from solid aerospace-grade COTS aluminum blocks (Al-7075-T73 or Al-6061-T6 series).

- **Surface robustness:** the standard chemical treatment is *hard anodizing*, optionally combined with local gold chemical conversion treatment (Alodine). This exponentially increases surface hardness and prevents molecular fatigue.
- **Cold-Welding:** the coating prevents zones where aluminum rubs during high acoustic stress inside the launcher fairing from experiencing vacuum cold-welding.

8.2.2 Geometry Standardization and CG Tolerance

By regulatory restrictions, the empirical modular sum of all subassemblies is formally recorded in the **Center of Gravity and Mass Geometric Report**.

- The stacked PC-104 distribution in a 2U requires a CG located within a few millimeters of the ideal geometric center on the 3 axes (< 2 cm tolerance) to prevent the vehicle from developing parasitic rotational impulses post-ejection, ensuring ADCS can damp residual polar moments in the first days.

8.3 Finite Element Analysis (FEA) Simulation

The parametric CAD model has yielded an optimized structure (lightening redundant material via cutouts on the main solar panel face), validated virtually:

- **Dynamic Resonance Modes:** theoretical verification of stiffness with the first natural resonance mode isolated above 100 Hz, decoupling the satellite chassis from destructive acoustics and natural frequencies historically transmitted by the inter-stage dynamic modulators of the injector rocket.
- **Static Stresses:** verification of longitudinal axial stress simulating extreme g-forces above > 10 g with a conservative elastic safety margin of 1.25 per GSFC-NASA standard recommendations.

Capítulo 9

Thermal Control

9.1 LEO Thermal Control Mechanisms

INTISAT's thermal control is transversal; it protects hardware to survive the extreme orbital cycles between severe direct solar illumination (+100 °C) and deep Earth-umbra eclipses (−40 °C). It is divided into two philosophies coupled to the spacecraft hardware:

9.1.1 Passive Control (Optical Coatings and Mass)

Inherently delegated to the construction of the *chassis (structure)*:

- **Radiators:** zones of non-anodized aluminum (bare or polished surfaces left at the end of geometric assembly), through which excess heat from the battery dissipates into deep space vacuum.
- **Insulation and optical coatings:** use of Kapton or aluminized tapes to internally shield PCB faces sensitive to the external cold bridge.

9.1.2 Active Control (EPS and OBC Integration)

To permanently protect the battery cells against low temperatures during eclipse (which would cause electrolyte freezing and capacity loss):

- **Flexible heaters:** resistive patches adhered directly to the *Battery Module*. Activation is orchestrated by the EPS control system through indirect BMS feedback and the 6 RTD temperature sensors of the central OBC architecture.

9.2 Thermal Test Campaign (AIT)

Behavior is simulated in CAD programs but rectified only empirically.

9.2.1 Thermal Cycling

The integrated satellite is subjected to dynamic chamber tests, cycling the temperature (e.g. between $-15\text{ }^{\circ}\text{C}$ and $+60\text{ }^{\circ}\text{C}$) repeated times (e.g. 10 cycles). The OBC must continue transmitting through the umbilical interface to detect thermal fatigue in critical perimeter solders.

9.2.2 Thermal Vacuum Test (Bake-Out LEO)

In sync with the launcher provider, the hardware is baked in high vacuum for weeks at $+60\text{ }^{\circ}\text{C}$ promoting maximum outgassing. This depuration is mandatory to certify that the satellite prevents contamination of the microscopy payload optics and neighboring instruments inside the launcher.

Capítulo 10

Ground Segment

10.1 INRAS Ground Operational Mission

The ground segment is the fundamental infrastructure for the pre-flight (AIT and FlatSat) and operational-in-flight phases. It directs the flow coordination from distributed Ground Stations to the Mission Control Center.

10.2 Ground Station Architecture, Control and Monitoring

In accordance with COTS architectures, the RF tracking circuit is the terrestrial mirror of the space TTC module:

- **Analog and inertial-tracking layer:** to maximize handshake on orbital passes (SSO or polar) to the millisecond, the station uses crossed *Quad-Yagi* arrays. They are governed by *Azimuth-Elevation automated rotators* that vectorially track ephemeris (TLE prediction files) updated by NORAD/Celestrak.
- **Amplifier base layer (LNA & Power Amp):** hosted close to the rotor to mitigate coaxial noise losses (NF). It amplifies the ultra-weak UHF footprint from space, while the power stage transmits between 50 and 100 W across the upper atmosphere to the CubeSat.
- **Digital and SDR layer:** extensive use of Software Defined Radios (SDRs). They continuously perform baseband digital demodulation, compensating for the extreme frequency stretching caused by the physical Doppler effect.

10.3 Logical Ground Operations Cycle (Ops)

Consolidated operations tasks interact directly responding to **OBC modes**:



- **Routine tracking:** passive cascaded decoding of *housekeeping* (vital for daily passive monitoring).
- **Scientific payload handling:** programmed sending of logical windows over territory, ordering the on-board computer to open and send the microscopy buffer to the CCSDS downlink.
- **Flight contingency resolution (HMAC/Security):** when notified via the Beacon flow that the satellite has automatically entered *Safe Mode*, the command line transmits asymmetric encrypted cryptograms (SHA/HMAC hashes), which manually and forcibly reactivate the microcontrollers (OTA reset), pulling the ADCS peripheral system out of *interlock* to save the mission.

Capítulo 11

Integration, Verification and Validation (AIT)

11.1 Overall Definition of the AIT Campaign

The Assembly, Integration and Testing (AIT) campaign establishes the mandatory set of experimental and systematic requirements before flight clearance (*Flight Model*). After this campaign, hardware and firmware are frozen (*design freeze*) to preserve qualification integrity.

Test profiles follow the standards **GSFC-STD-7000 (GEVS)**, **ECSS-E-ST-10-03** (testing), and **CubeSat Design Specification Rev 14** (Cal Poly).

11.2 Assembly Instructions and FlatSat

Prior to final physical integration, key tests are performed at the expanded-system level.

11.2.1 FlatSat (Bench Tests)

The *FlatSat* configuration consists of interconnecting all satellite boards outside the chassis, via jumper cables, to perform *Hardware-In-the-Loop* (HIL) tests and identify interface issues before mechanical assembly.

FlatSat sequence:

1. Board interconnection on bench per PC-104 topology.
2. Continuity check and power rails verification (3.3 V, 5 V, V_{batt}).
3. Controlled power-up with a *current-limited* lab power supply.
4. Verification of *boot* and handshake OBC $\leftrightarrow$ EPS $\leftrightarrow$ UHF.
5. Execution of typical commands (housekeeping, beacon, payload trigger).

6. Validation of the four OBC operating modes: Initialization, Nominal, Safe, De-commissioning.

11.2.2 Mechanical Integration Procedure

1. Surface cleaning and dimensional verification of the structure.
2. Mounting of boards on the PC-104 bus with standardized torques (TBD N·m per M3 screw).
3. Internal harness connection per wiring diagram.
4. Mounting of solar panels and folded antennas.
5. Insertion of kill-switches and continuity check of RBF (*Remove-Before-Flight*).
6. Final structural closure and verification of the 2U CubeSat envelope.

11.3 Physical and Environmental Validation Test Plan

Tests qualify physical and functional properties of the satellite against launch and LEO operation environments.

11.3.1 Physical-Mechanical Tests

- **Dimensions:** CMM or 3D-scan measurement against the 2U envelope. Tolerance: $\pm 0,1$ mm on rails.
- **Physical interface verification (Fit Check):** insertion/extraction into the P-POD injector. Verify clearances and kill-switch orientation.
- **Mass and Center of Gravity (CG):** precision balance and suspension method. CG must lie within $< 2,0$ cm of geometric center on the 3 axes (CDS Rev 14).
- **Sinusoidal and Random Vibration (Vibration & Shock):** vibration table on 3 axes (X, Y, Z) with launcher-representative profile (TBD: Falcon 9, PSLV, Soyuz, etc.).
 - Sinusoidal *sweep*: 5–100 Hz, ± 2 mm or 2.5 g.
 - *Random*: *Component Minimum Workmanship* profile from GSFC-STD-7000.
- **Shock:** sinusoidal or simulated pyrotechnic, ~ 1000 g peak, 0.5 ms.

11.3.2 Space Environment and EMC Tests

- **Thermal Cycling:** chamber -15 to $+60$ °C, minimum 10 cycles. The OBC must keep transmitting via umbilical to detect thermal fatigue in critical solders.
- **Thermal Vacuum (TVAC + Bake-Out):** $+60$ °C in high vacuum for weeks, promoting outgassing and certifying cleanliness for launcher integration.
- **Electromagnetic Compatibility (EMC):** immunity and intermodulation testing between subsystems, with emphasis on the UHF/VHF receiver to rule out self-interference.

11.3.3 Recommended AIT campaign sequence

Based on the FloripaSat-2 practice and CDS Rev 14:

1. Visual inspection and mass/CG verification.
2. Fit check in P-POD.
3. Vibration (sequential X, Y, Z).
4. Shock.
5. Visual and electrical re-inspection.
6. Thermal cycling.
7. TVAC + Bake-Out.
8. EMC.
9. Final functional test (FlatSat-equivalent on integrated satellite).

Capítulo 12

Mission and Project Management

12.1 Systems Management and Milestones

The design and launch of a CubeSat 2U platform such as INTISAT is formally articulated under the V-Model of space systems engineering (NASA-SP-2007-6105, ECSS-E-ST-10C). The trajectory follows a strict path: SRR (*System Requirements Review*) → PDR (*Preliminary Design Review*) → **CDR (Critical Design Review)** of the engineering model → AIT (*Assembly, Integration and Testing*) → FRR (*Flight Readiness Review*) → Flight Model. This document corresponds to the engineering-model CDR.

12.2 Schedule, WBS and Coupled Risk Management

- **Work Breakdown Structure (WBS):** subdivides and parallelizes operational requirements and manufacturing loads into a hierarchical matrix scheme. Branched engineering threads (e.g. *OBC software*, *EPS thermal manufacturing*, *AI payload training*) progress encapsulated and converge at transversal mechanical milestones (e.g. harness integration and FlatSat tests).
- **Technical Risk Control and Mitigation Matrix:** stochastic foresight tied to hardware failure control. It catalogues potential on-orbit disasters (e.g. radiation-induced short-circuits, BGA solder failures after launcher vibration).

It also outlines technical pre-flight responses to inevitable contingencies: for example, in case of severe under-voltage due to persistent eclipse, planning delegates by-design preventive intelligence to the *Safe Mode* reaction, stopping fatal degradation, disconnecting collateral instruments and sending emergency requests to the global ground network awaiting troubleshooting on console.

Apéndice A

References

Standards and manuals

- NASA-SP-2007-6105 Rev2 — *NASA Systems Engineering Handbook*, 2007.
- ECSS-E-ST-10C Rev1 — *Space Engineering: System Engineering General Requirements*, ESA, 2017.
- ECSS-E-ST-10-02C Rev1 — *Verification*, ESA, 2018.
- ECSS-E-ST-10-03C Rev1 — *Testing*, ESA, 2022.
- ECSS-E-ST-20C Rev2 — *Electrical and electronic*, ESA, 2022.
- ECSS-E-ST-31C — *Thermal control general requirements*, ESA, 2008.
- ECSS-M-ST-40C Rev1 — *Project Management*, ESA, 2008.
- GSFC-STD-7000A — *General Environmental Verification Standard (GEVS)*, NASA Goddard, 2019.
- CubeSat Design Specification Rev 14 — Cal Poly SLO, 2020.
- CCSDS 133.0-B-2 — *Space Packet Protocol*, CCSDS, 2020.
- ISO 21502:2020 — *Project, programme and portfolio management*, ISO.
- PMBOK Guide, 7th Edition — PMI, 2021.

Technical reference documents

- Wertz, J. R., Everett, D. F., & Puschell, J. J. — *Space Mission Engineering: The New SMAD*, Microcosm Press, 2011.
- Larson, W. J. & Wertz, J. R. — *Space Mission Analysis and Design*, 3rd ed., 1999.



- SpaceLab UFSC — *GOLDS-UFSC (FloripaSat-2) Design and Mission Overview*, June 2023. Available at <https://github.com/spacelab-ufsc/floripasat2-doc>.

Internal INTISAT documents (this repository)

- 03_recursos/APSCO/APSCO_Invitacion_Summer_Camp_Xian_2026.pdf
- 03_recursos/APSCO/GUIA_estandares_y_referencias.md
- 03_recursos/presupuestos/Requerimiento de energía - INTISAT.xlsx
- 01_CDRs_subistemas/ — Detailed CDRs per subsystem (OBC, UHF, EPS, Microscopy).

Apéndice B

Annexes

B.1 Additional diagrams, tables and budgets